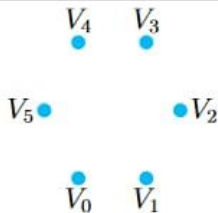
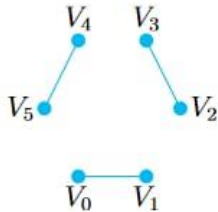
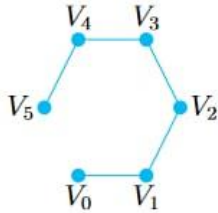
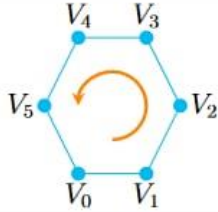
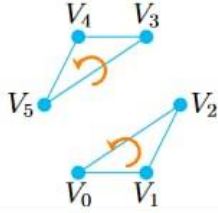
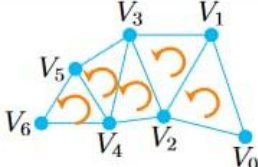
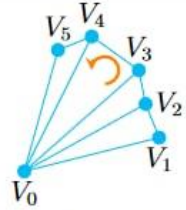
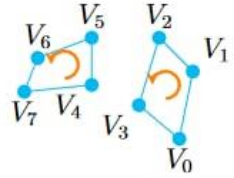
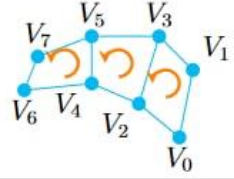
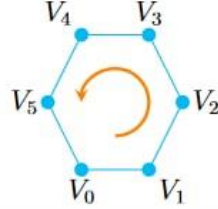


5.3 Geometric Drawing Primitives

Primitive	Description	Sample
GL_POINTS	individual points	
GL_LINES	pairs of vertices interpreted as individual line	
GL_LINE_STRIP	series of connected line segments	
GL_LINE_LOOP	same as above, with a segment added between last and first vertices	
GL_TRIANGLES	triples of vertices interpreted as triangles	
GL_TRIANGLES_STRIP	triples of vertices interpreted as triangles	

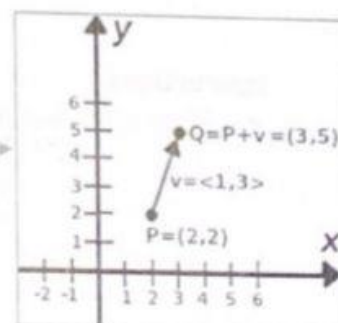
Primitive	Description	Sample
GL_TRIANGLES_FAN	linked fan of triangles	
GL_QUADS	Quadruples of vertices interpreted as four-sided polygons	
GL_QUAD_STRIP	linked strip of quadruples	
GL_POLYGON	boundary of a simple, convex polygon	

مهم تفهم كيف تسوي الاشكال اللي فوق
تحت تكملة الاختبار

Q1. Select the correct answer.

marks)

- 1.1. Raster graphics are composed of
- Pixels
 - Paths
 - Palette
 - None of these
- 1.2. GUI stands for -
- Graphics Uniform Interaction
 - Graphical User Interaction
 - Graphical User Interface
 - None of the above
- 1.3. The RGB coloring model is:
- a subtractive color
 - an additive color
 - subtractive and additive color
 - none of the above
- 1.4. The following package(s) is(are) needed to develop graphics framework using OpenGL with Python:
- Pygame
 - Numpy
 - PyOpenGL
 - All the above
- 1.5. Generate two-dimensional images of three-dimensional scenes is called _____:
- Rendering
 - Shading
 - Zooming
 - None of the above
- 1.6. OpenGL is _____
- Graphics Application Programming Interface (API)
 - Graphics editing environment
 - Programming Language
 - None of the above
- 1.7. Vector graphics contain:
- Grid of pixels
 - Mathematical formulas
 - Software instructions
 - None of the above
- 1.8. The shown figure represents:
- Vector addition
 - vector rotation
 - Vector scalar
 - None of the above



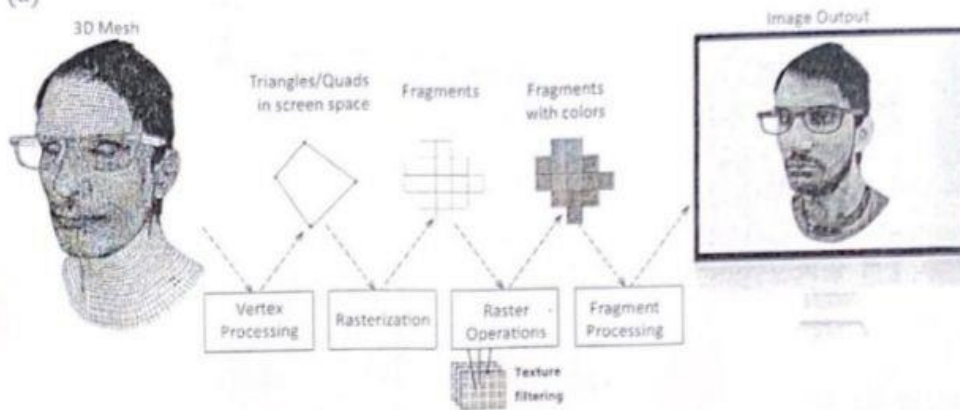
Q2. Answer the following questions with True (T) or False (F)

(4 marks)

- a. The color depth is the number of bits used to indicate the color of a single pixel
- b. HSV coloring mode is a different way to manipulate the RGB colors
- c. The number of pixels that need to be computed in each second to attain the rate of 60 FPS on a 1920 x 1080 screen = $1920 \times 1080 \times 60$
- d. The function `glPointSize(5)` is used to define the width of point 5 pixels

Q3. Choose one of the following figures and explain in detail what graphics principle do they reflect. (4 marks)

(a)



(b)

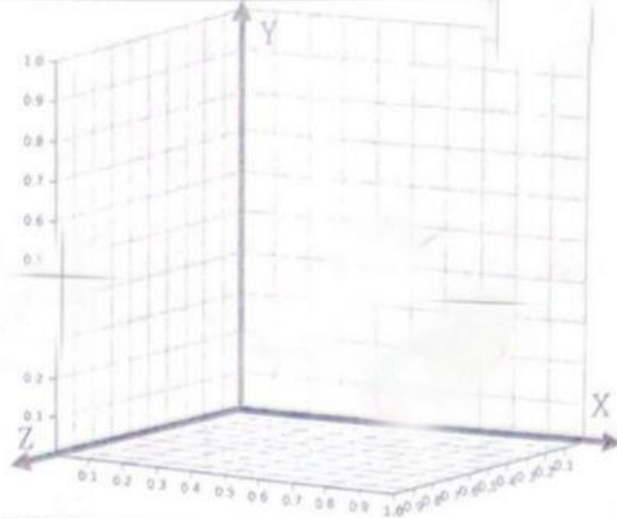


Q4. Draw the output of running the following OpenGL functions (/6 marks)

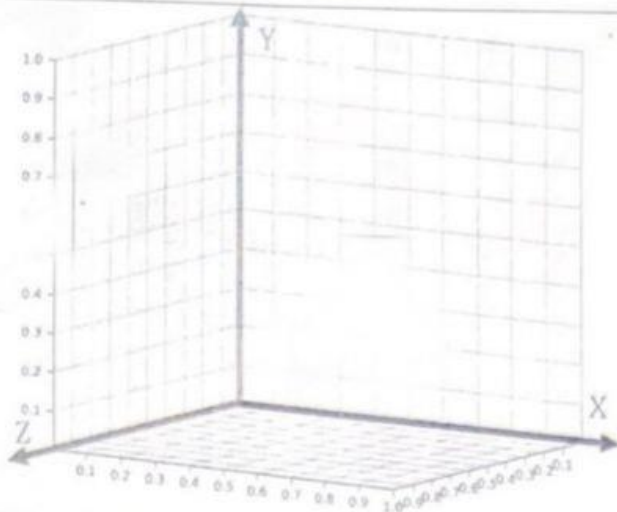
Use the following position values for each vertex in (x, y, z) coordinate system to draw the output of the functions indicated in A, B & C.

$V1 = (0.1, 0.1, 0.0)$, $V2 = (0.4, 0.4, 0.0)$, $V3 = (0.6, 0.2, 0.0)$, $V4 = (0.4, 0.1, 0.0)$

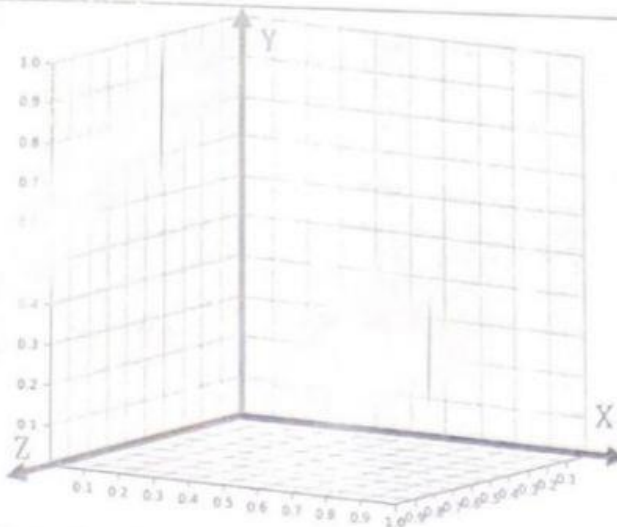
(A) GL_LINES



(B) GL_LINE_STRIP

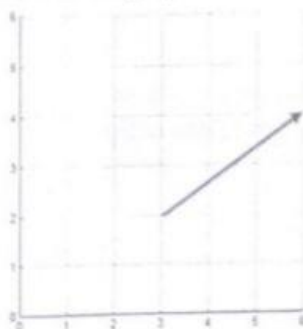


(C) GL_LINE_LOOP



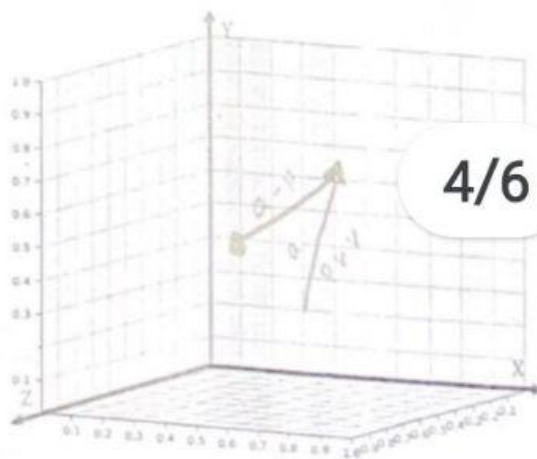
/8)

- a. 2V
b. 0.5 V
c. -V



- Assume that F is a linear function, $F(i) = \langle 1, 2 \rangle$ and $F(j) = \langle 3, 1 \rangle$.

- Represent the following:
 $Q = P + V$ and $V = Q - P$

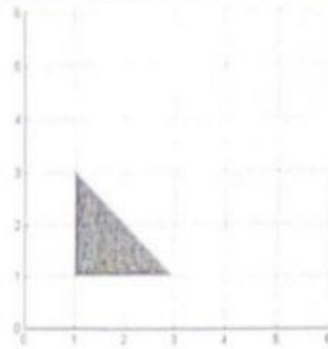


Q6. Process the operations for the following

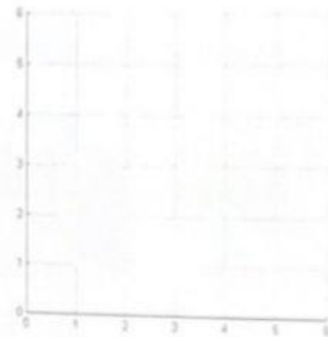
(/8)

The given triangle represented in 2D coordinate system

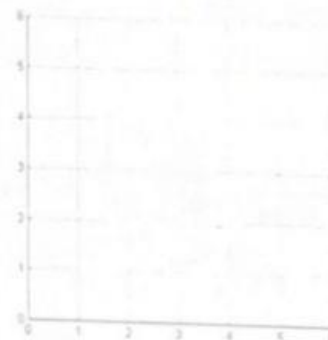
Draw the triangle transforming output when the following 4x4 matrices or OpenGL function are provided.



a)
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



b) `glTranslatef (1.0, 1.0, 0.0)`



$$c) \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



d) glScalef (2.0, 2.0, 0.0)



MAFIA FCIT20:

<https://t.me/FCIT20Male>